

NeuroScope

AI-Powered 3D Neural Network Architecture Visualizer and Analyzer

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The Problem We Solve

Every year, millions of students across Africa and the world begin their journey into deep learning and machine learning. They enroll in online courses, attend university programs, and follow tutorials that teach them to write code like `model.fit()` and achieve high accuracy scores. But beneath this surface-level success lies a fundamental gap: **most students do not understand what happens inside the models they build.**

They copy architectures from GitHub repositories without knowing why a ResNet uses skip connections, or why a transformer uses multi-head attention, or why adding batch normalization after a convolutional layer stabilizes training. When their model fails to converge, they have no way to diagnose the problem. When they see a new architecture for the first time, they cannot read the code and understand what each block does. The model is a black box - they can operate it, but they cannot see inside it.

This problem is especially acute in Africa. Across the continent, universities are rapidly expanding their computer science and data science programs. Egypt's Ministry of Communications and Information Technology has made digital transformation a national priority. Nigeria, Kenya, South Africa, Rwanda, and Tunisia are investing heavily in AI education. The African Union's Agenda 2063 and the STISA 2034 strategy call for a new generation of technically skilled youth who can lead Africa's digital industrialization.

But the tools available to these students were not designed for them. Netron shows static two-dimensional diagrams with no explanation. TensorBoard requires running training before visualization works and is locked to the TensorFlow framework. TensorSpace.js, which once offered three-dimensional visualization, has been abandoned since 2019. None of these tools explain what a layer does. None of them detect architectural mistakes. None of them work offline on a student's phone in a rural area with limited internet connectivity. And none of them connect the visualization back to the student's actual code.

NeuroScope changes this.
Our Solution

NeuroScope is a web-based tool and Visual Studio Code extension that transforms how students learn deep learning by making neural network architectures visible, interactive, and understandable.

A student uploads their model file - whether it is an ONNX, PyTorch, Keras, or TensorFlow Lite file - or simply opens their Python script or Jupyter notebook. NeuroScope parses the code or model file and generates a **three-dimensional interactive visualization** of the architecture. Each layer type is represented by a distinct three-dimensional shape: convolutional layers appear as boxes representing feature map volumes, fully connected layers appear as flat planes representing weight matrices, recurrent layers appear as cylinders representing cyclical data flow, and attention layers appear as octahedrons representing multi-head patterns.

The student clicks any three-dimensional component and immediately sees three things: a detailed description of what that layer does in plain language, the exact code block from their script or notebook that created that layer, and the layer's parameters including input shapes, output shapes, parameter count, floating-point operations, and

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memory footprint. This bidirectional mapping between code and visualization means that a student who has never seen a transformer architecture before can read the code and understand it by exploring the three-dimensional model, and a student who understands the architecture can locate and debug the corresponding code.

The architecture health check engine analyzes the model and detects over forty-seven common anti-patterns organized into four categories: layer-level issues such as missing activation functions or sigmoid in deep networks, architecture-level issues such as missing skip connections or parameter explosion in fully connected layers, efficiency issues such as redundant convolutional layers or unnecessarily large kernels, and task-specific issues for convolutional networks, recurrent networks, transformers, and autoencoders. Each finding includes a severity level, a clear explanation of why it is a problem, and a specific suggested fix. It is like a spell-checker for neural network design.

The real-time simulation means that when a student edits a parameter in their code - changing the number of filters in a convolutional layer, adjusting the dropout rate, or adding a new layer - the three-dimensional visualization updates instantly before the student runs the code. This live preview allows students to experiment with architectural decisions and see their consequences immediately, developing intuition that would otherwise require months of trial and error.

When the student runs their code, NeuroScope animates the forward pass: data flows through the network as animated particles, tensor shapes change visually at each layer, and the student watches their input transform into a prediction in real time. This simulation makes the abstract mathematics of backpropagation and gradient descent tangible and visible.

Impact for Africa and the World

NeuroScope is free, open-source, and designed for the environments where it is needed most. It works entirely in the browser with no installation required. It functions as a progressive web application that caches its assets after the first load and works offline - critical for students in areas with intermittent internet connectivity. It requires no GPU, no cloud computing credits, and no expensive software licenses. It runs on a smartphone, a tablet, or a ten-year-old laptop.

The tool supports five languages: English, French, Arabic, Swahili, and Portuguese - covering the majority of African students who are excluded by English-only tools. Egypt's twenty-two Arabic-speaking governorates, Francophone West and Central Africa, East Africa's Swahili-speaking regions, and Lusophone Southern Africa can all use NeuroScope in their native language.

NeuroScope aligns directly with the African Union's STISA 2034 priority of building scientific and technological literacy, with Agenda 2063's call for inclusive and sustainable development, and with Egypt's national strategy for digital transformation in education. It addresses the mentorship gap that prevents talented African students from reaching their potential: when no senior machine learning engineer is available to review a student's architecture, NeuroScope provides that review automatically, clearly, and patiently.

But this tool is not only for Africa. Every computer science student in the world who struggles to understand why their convolutional neural network is not learning, or what each block in a transformer does, or how to diagnose a vanishing gradient problem, will benefit from NeuroScope. The global machine learning community needs better educational tools, and NeuroScope is built to serve that need.

Sustainability

NeuroScope is released under the MIT license as a fully open-source project. Its development model is community-driven: university professors can integrate it into their curricula, students can contribute new analysis rules and language translations, and institutions can deploy their own instances. The tool is designed to grow with the community that uses it.

We are building NeuroScope not just as a competition entry, but as lasting infrastructure for machine learning education in Africa and beyond.

Word Count: ~800

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